ASSIGNMENT 1

Research Paper Re-Implementation

1. Assignment Task

Select ONE research article from the provided attached folder and conduct a complete research paper reproduction and implementation study. Read and understand the selected work, assess its feasibility and reproducibility, independently re-implement the proposed methodology, reproduce the reported experiments, and critically explain similarities or differences between the published and obtained results.

2. Important: Original Implementation Requirement

You must select an article whose complete implementation/source code is not publicly available. If the complete source code or implementation of the selected article is publicly available, that article should NOT be selected for this assignment. The core methodology must be independently implemented based on the student's understanding of the paper. Copying, substantially reproducing, or modifying publicly available source code and presenting it as one's own work will be considered academic plagiarism and will be dealt with according to the applicable plagiarism/academic integrity policy. Standard libraries, frameworks, APIs, and tools may be used, but the core methodology must be independently implemented. Students are responsible for checking source-code availability before selecting an article.

3. Research Paper Analysis and Feasibility

After carefully reading the selected article, answer the following:

1. What problem does the research work address, and what are its main objectives and contributions?
2. What methodology/approach has been proposed?
3. What dataset(s) are used, and are they publicly available?
4. What programming language, libraries, frameworks, tools, and computational resources are required?
5. Is the proposed methodology practically feasible to implement with the available resources?
6. Are sufficient implementation details provided to reproduce the research?
7. What information is missing, unclear, or difficult to reproduce?

4. Independent Re-Implementation

Independently re-implement the methodology proposed in the selected article. Follow the original methodology, dataset, preprocessing, experimental settings, and parameters as closely as possible. Clearly document every assumption, modification, approximation, or deviation made during implementation. Submit complete, executable source code with clear execution instructions.

5. Experimental Reproduction

Run the implemented system and compare your results with those reported in the original article. Use appropriate evaluation metrics and, where applicable, perform multiple runs.

Evaluation Metric	Published Result	Your Result	Difference
Metric 1			
Metric 2			
Metric 3			
Metric 4			

6. Critical Analysis of Results

Clearly determine whether your implementation produced (A) comparable results, (B) improved results, or (C) lower results than the original research. Provide technically valid justification.

- Dataset version, availability, split, and preprocessing differences
- Experimental settings, parameters, and configuration differences
- Randomness, random seeds, or stochastic behavior
- Software/library/framework versions and computational resources
- Missing or ambiguous implementation details in the original paper
- Any changes or deviations introduced during implementation

A higher numerical result alone does not constitute an improvement, and a lower result alone does not constitute an unsuccessful reproduction. Conclusions must be supported by appropriate experiments, fair comparison, and valid technical arguments.

7. IEEE Technical Research Report

Prepare a technical research paper in IEEE format containing:

8. Abstract
9. Introduction
10. Research Problem and Selected Article
11. Methodology
12. Dataset and Preprocessing
13. Implementation
14. Experimental Setup
15. Results and Comparison
16. Critical Analysis and Discussion
17. Limitations
18. Conclusion
19. References

The report must clearly distinguish the original research methodology, your independent implementation, and any modifications introduced by you. All references must follow IEEE referencing style.

8. Code Submission

- Complete source code
- README / execution instructions
- Required libraries and packages
- Dataset information/source
- Configuration and parameters
- Experimental results
- Graphs/plots where applicable
- Any additional files required to execute the implementation

9. Viva Voce

Evaluation will include a Viva Voce. Students must be able to explain the selected article, dataset, methodology, implementation decisions, experimental setup, published results, their own results, differences between the results, technical justification, and submitted source code.

Evaluation Rubric

Evaluation Component	Description	Marks
Research Paper Understanding & Analysis	Problem, objectives, methodology, and contributions.	15
Feasibility & Reproducibility Assessment	Dataset, resources, implementation requirements, and reproducibility analysis.	10
Independent Implementation	Correctness and completeness of the re-implemented methodology.	20
Code Quality & Documentation	Organization, readability, README, requirements, and execution instructions.	10
Experimental Reproduction & Results	Experimental setup, comparison, and appropriate evaluation.	15
Critical Analysis & Technical Justification	Evidence-based explanation of comparable, improved, or lower results.	10
IEEE Technical Report	IEEE format, technical content, writing, figures/tables,	10

	discussion, references.	
Viva Voce	Technical understanding, code defense, and explanation of findings.	10
Total		100

Submission Requirements

- Submit: (1) complete source code and supporting files
- IEEE-format technical research paper
- Participate in the Viva Voce.